Discrete Calculus & Number Theory Problem Set

30 Problems — Multiple Choice

1. $I = \int_0^{\infty} x/(e^x-1)\,dx$.
Without evaluating I directly by antiderivatives, determine its exact value.
A) $\pi^2/12$
B) $\pi^2/6$
C) $\pi^2/3$
D) $2\pi^2/3$
E) $\pi^3/6$

2. Define $S = \sum_{k=1}^{99} 1/(k(k+1)(k+2))$.
Which expression is equal to S?

A) 1/198
B) 1/100
C) $1/6 - 1/(100\cdot101)$
D) 1/4
E) 1/3

3. Evaluate $I = \int_0^{\pi/2} \ln(\sin x)\,dx$.
A) $-\pi \ln 2$
B) $-(\pi/2)\ln 2$
C) $-\pi/2$
D) $-\pi$
E) 0

4. Let p be a prime greater than 3. Prove mentally enough structure to determine the remainder of $(p^2-1)/24$ modulo p.
Which answer is necessarily correct?

A) 0
B) 1/24
C) −1/24
D) (p−1)/24
E) It depends on p

5. Evaluate the double integral over the region R:
$R = \{(x,y): x^2+y^2 \le 1, x+y \ge 0\}$
$I = \iint_R (x^2+y^2)\,dA$.
A) π/8
B) π/4
C) π/2

D) $2\pi/3$

E) π

6. Let $I = \int_0^1 [\ln(1+x)]/[1+x^2]\,dx$.
Which of the following is the most useful first substitution for reducing I to a symmetry identity?

A) $x = \tan\theta$
B) $x = e^t$
C) $x = (1-t)/(1+t)$
D) $x = \sin\theta$
E) $x = t^2$

7. Let $a_1=1$ and $a_{n+1} = a_n + \gcd(a_n,n)$ for $n\ge1$.
Which statement must be true?

A) a_n is always prime
B) a_n is always odd
C) $a_n \ge n$
D) a_n is eventually constant
E) a_n is divisible by n for every n

8. Evaluate exactly: $I = \int_0^1 [\ln x \cdot \ln(1-x)]/x\,dx$.
A) $\pi^2/6$
B) $\pi^4/90$
C) $\pi^4/36$
D) $2\zeta(3)$
E) $\pi^3/12$

9. Let $F(a) = \int_0^\infty e^{-x}\cos(ax)\,dx$.
Without using a table of Laplace transforms, determine $F''(0)$.

A) $-1/2$
B) -1
C) 0
D) 1
E) 1/2

10. Find the exact value of $I = \int_0^1 \int_0^1 1/(1+xy)\,dx\,dy$.
A) $\ln 2$
B) $\pi^2/12$
C) $\pi^2/6$
D) $2\ln 2$
E) 1

11. Let n be a positive integer satisfying
$n \equiv 2 \pmod 3$, $n \equiv 3 \pmod 5$, $n \equiv 4 \pmod 7$.
What is the least positive value of n?

A) 17

B) 23

C) 53

D) 68

E) 83

12. Evaluate the triple integral $I = \iiint_V (x^2+y^2+z^2)\, dV$,
where $V = \{(x,y,z): x^2+y^2+z^2 \le 1, x+y+z \ge 0\}$.

A) $\pi/5$

B) $2\pi/15$

C) $4\pi/15$

D) $2\pi/5$

E) $4\pi/5$

13. Let $I_n = \int_0^{\pi/2} \sin^n x\, dx$.
Instead of computing I_n recursively, determine which quantity is independent of n: A) I_n/I_{n+1}

B) $(n+1)I_n/I_{n+2}$

C) $(n+2)I_n/I_{n+2}$

D) nI_n/I_{n-2}

E) $I_n - I_{n+1}$

14. Determine the exact value of $I = \int_0^1 [\arctan x]/[1+x^2]\, dx$. A) $\pi^2/16$

B) $\pi^2/12$

C) $\pi^2/8$

D) $\pi/4$

E) 1

15. Let $P(x) = x^{100} + x^{99} + \cdots + x + 1$.
Without expanding or finding its roots individually, determine the remainder when $x^{100}P(1/x) - P(x)$ is divided by $x-1$.

A) -100

B) -50

C) 0

D) 50

E) 100

16. Evaluate $I = \int_0^{\infty} dx/(1+x^4)$.

A) $\pi/4$

B) $\pi/(2\sqrt{2})$

C) $\pi/\sqrt{2}$

D) $\pi/2$

E) $\sqrt{2}\,\pi/4$

17. Let S be the set of integers k with $1 \le k \le 2026$ for which k^2+k+1 is divisible by 7. How many elements are in S?

A) 288

B) 289

C) 290

D) 291

E) 292

18. Evaluate the iterated integral $I = \int_0^1 \int_0^1 [1/(1+x^2y^2)]\, dy\, dx$. A) $\pi^2/16$

B) $\pi^2/12$

C) G (Catalan's constant)

D) ln 2

E) $\pi/4$

19. Suppose f is twice differentiable and satisfies $f(x)+f(1-x)=x^2$ for every real x. What is $f''(1/2)$?

A) −2

B) −1

C) 0

D) 1

E) 2

20. Evaluate exactly: $I = \int_0^1 [x^{-1/2}-1]/\ln x\, dx$.

A) ln 2

B) ln(3/2)

C) 2 ln 2

D) $\pi/2$

E) 1

21. Let p be a prime greater than 3. Which expression is always an integer?

A) $(p-1)/6$
B) $(p^2-1)/12$
C) $(p^3-p)/12$
D) $(p^2+1)/12$
E) $(p+1)/12$

22. Consider $I = \int_0^{\pi/2} [x/(\sin x + \cos x)]\, dx$.
A symmetry argument gives the value without finding an antiderivative. What is I? A) $\pi^2/16$
B) $\pi^2/12$
C) $\pi^2/8$
D) $\pi/4$
E) $\pi/2$

23. Evaluate $I = \iint_{[0,1]^2} \ln(1-xy)\, dA$.
A) $-\pi^2/12$
B) $-\pi^2/6$
C) $-\zeta(3)$
D) $-\pi^2/6 + 1$
E) $-2\zeta(3)$

24. Let $S_n = \sum_{k=1}^{n} 1/(k^2+k)$. Determine $\lim_{n\to\infty} S_n$.
A) 1
B) $\pi^2/6$
C) 3/2
D) 2
E) Diverges

25. Evaluate $I = \int_0^1 \int_0^1 \int_0^1 1/(1+xyz)\, dx\, dy\, dz$.
A) $\pi^2/12$
B) $\zeta(3)$
C) $\pi^4/90$
D) $2\zeta(3)$
E) $\ln 2$

26. Evaluate $I = \int_0^1 \ln(1+x)/x\, dx$.
A) $\pi^2/12$
B) $\pi^2/6$
C) $\pi^2/8$
D) $(\ln 2)^2$
E) $\pi^2/4$

27. Without listing them by hand, determine the number of positive divisors of 2025 that are perfect squares.

A) 4

B) 5

C) 6

D) 7

E) 8

28. Evaluate $I = \int_0^{\infty} (\sin x)/x \, dx$.

A) $\pi/4$

B) $\pi/2$

C) π

D) It diverges

E) 0

29. Using Fermat's Little Theorem rather than direct computation, find the remainder when 7^{100} is divided by 13.

A) 1

B) 3

C) 9

D) 10

E) 12

30. Let $S = \sum_{k=1}^{\infty} 1/(4k^2-1)$.
Determine the exact value of S.

A) 1/4

B) 1/2

C) 1

D) ln 2

E) $\pi/4$